JATIN DESWAL

Noida, India | 9818049435 | jatindeswal28@gmail.com | linkedin.com/in/jatindeswal | github.com/Jatindeswal

SUMMARY

AI infrastructure builder and full-stack software engineer skilled in AI agents, RAG pipelines, and network observability. Experienced across the stack — Python/JavaScript backends, REST APIs, databases, and interactive front-ends — with a focus on automation, root cause analysis, and building reliable, production-grade systems that turn data into insight.

EDUCATION

Amity University 2023 – 2027
B.Tech, Computer Science & Engineering CGPA: 7.51 / 10
Relevant Coursework: Data Structures & Algorithms, Computer Networks, Operating Systems, Database Management Systems, Artificial Intelligence & Machine Learning

Scholars Rosary Senior Secondary School (CBSE)
Class XII — 87.20% | Class X — 87.33%

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

AI/ML & Agents: LLM Orchestration (LiteLLM), LangGraph, RAG Pipelines, Vector Databases (ChromaDB), ReAct Agent Loops, Scikit-Learn, Vision-Language Models (VLMs)

Full-Stack Development: REST APIs, Flask, React/TypeScript, D3.js, Server-Sent Events (SSE), DOM Traversal, Chrome Manifest V3

Cloud & Infrastructure: AWS (IAM, EC2, VPC), Network Packet Analysis (Scapy), Real-Time Observability, Anomaly Detection, SQLite, APScheduler

Tools & Platforms: Git/GitHub, Linux, Unit Testing & Test Automation

Concepts: Automation, Root Cause Analysis, ITSM Fundamentals, Multi-LLM Routing

PROJECTS

HiveCore — Local-First AI Agent Framework *Python, LiteLLM, ChromaDB, APScheduler, React/TypeScript*
github.com/Jatindeswal/hive

- Engineered a privacy-first AI agent runtime supporting 100+ LLM providers via LiteLLM, enabling flexible multi-model routing and reducing vendor lock-in for downstream automation tasks.
- Built a hybrid retrieval system (BM25 + vector similarity via Reciprocal Rank Fusion) over a tiered ChromaDB/SQLite memory store to improve long-term reasoning accuracy.
- Automated recurring operational tasks with a cron-based scheduler (APScheduler) and a self-reflective ReAct (Reason→Act→Observe) agent loop.
- Ensured production reliability by writing 352 unit tests achieving 80.84% coverage, supporting stable extensibility via hot-reloaded, sandboxed Python skills.

Snifferon — Advanced Network Visualization & Observability Suite *Python, Flask, Scapy, Scikit-Learn, D3.js*
github.com/Jatindeswal/Snifferon

- Built a real-time full-stack observability dashboard (Flask backend, D3.js front-end) that captures, enriches, and visualizes live network traffic (HTTP/TLS/DNS) to accelerate root cause analysis.
- Reduced anomaly detection time by integrating an ML-powered scoring engine (Scikit-Learn) with behavioral clustering to surface suspicious traffic patterns.
- Designed an interactive D3.js force-directed graph with protocol filtering, geographic traffic mapping, and drill-down node inspection.
- Validated production-readiness through a 154-test suite across 10 modules and persistent historical baselining via SQLite-backed hourly metrics.

The Eye Opener — AI Fact-Checking RAG Pipeline *Python, Flask, LangGraph, ChromaDB*
github.com/Jatindeswal/eye-opener

- Designed a multi-stage LangGraph orchestration pipeline (preprocessor → surgeon → diver → skeptic → scorer) to automate claim extraction, evidence retrieval, and verification.
- Increased factual accuracy by combining local-first ChromaDB retrieval with live web enrichment and a skeptic stage that flags weak sourcing.
- Delivered a responsive UX by streaming multi-stage pipeline progress to the browser in real time using Server-Sent Events (SSE).
- Architected a fallback LLM chain (Ollama → Cerebras/Groq) to maintain uptime under varying local and cloud availability.

Auto Captcha Solver — Browser Automation Extension *JavaScript, Chrome Manifest V3, Vision-Language Models*
github.com/Jatindeswal

- Built a Chrome extension (Manifest V3) that scans page DOM structures to auto-detect and resolve CAPTCHAs, integrating swappable multimodal AI providers (Gemini, GPT-4o, Claude) with API calls isolated in background service workers for security.